```
    d_hidden = error_hidden * sigmoid_derivative(hidden_output)

    # ---------- Update Weights and Biases ----------
    W2 += hidden_output.T.dot(d_output) * learning_rate
    b2 += np.sum(d_output, axis=0, keepdims=True) * learning_rate

    W1 += X.T.dot(d_hidden) * learning_rate
    b1 += np.sum(d_hidden, axis=0, keepdims=True) * learning_rate

    if epoch % 1000 == 0:
        print(f"Epoch {epoch}, Loss: {loss:.6f}")

print("\nFinal Predictions:")
for i in range(len(X)):
    prediction = final_output[i][0]
    predicted_class = 1 if prediction >= 0.5 else 0
    print(f"Input: {X[i]} -> Predicted: {prediction:.4f} -> Class:
{predicted_class}")
```

```
Epoch 0, Loss: 0.294351
Epoch 1000, Loss: 0.244410
Epoch 2000, Loss: 0.203539
Epoch 3000, Loss: 0.153349
Epoch 4000, Loss: 0.046331
Epoch 5000, Loss: 0.015615
Epoch 6000, Loss: 0.008448
Epoch 7000, Loss: 0.005614
Epoch 8000, Loss: 0.004146
Epoch 9000, Loss: 0.003264

Final Predictions:
Input: [0 0] -> Predicted: 0.0540 -> Class: 0
Input: [0 1] -> Predicted: 0.9505 -> Class: 1
Input: [1 0] -> Predicted: 0.9501 -> Class: 1
Input: [1 1] -> Predicted: 0.0536 -> Class: 0
```

LAB 8| 12 March 2026

Change the batch size

```
import tensorflow as tf
from tensorflow.keras.datasets import mnist
from tensorflow.keras.utils import to_categorical
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense
import numpy as np

# Load dataset
```

```python
(X_train, y_train), (X_test, y_test) = mnist.load_data()

# Normalize
X_train = X_train / 255.0
X_test = X_test / 255.0

# Flatten
X_train = X_train.reshape(-1, 784)
X_test = X_test.reshape(-1, 784)

# One-hot encoding
y_train = to_categorical(y_train, 10)
y_test = to_categorical(y_test, 10)

# Different batch sizes
batch_sizes = [32, 64, 128, 256]

for batch in batch_sizes:
    print("\nTraining with batch size:", batch)

    # Model
    model = Sequential([
        Dense(10, activation='softmax', input_shape=(784,))
    ])

    model.compile(optimizer='adam',
                  loss='categorical_crossentropy',
                  metrics=['accuracy'])

    model.summary()

    # Train model
    model.fit(X_train, y_train, epochs=15, batch_size=batch)

    # Evaluate
    test_loss, test_acc = model.evaluate(X_test, y_test)
    print("Test accuracy with batch size", batch, ":", test_acc)
```

```
Training with batch size: 32

Model: "sequential_12"
```

```
┏━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━┳━━━━━━━━━━━━━━━━━━━━━━━━━━━━━┳━━━━━━━━━━━━━━━━━
━━━┓
┃ Layer (type)                         ┃ Output Shape                ┃
Param # ┃
┡━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━╇━━━━━━━━━━━━━━━━━━━━━━━━━━━━━╇━━━━━━━━━━━━━━━━━
━━━┩
│ dense_12 (Dense)                     │ (None, 10)                  │
7,850 │
```

```
└─────────────────────────────────────┴─────────────────────────────┴──────────
──────┘

 Total params: 7,850 (30.66 KB)

 Trainable params: 7,850 (30.66 KB)

 Non-trainable params: 0 (0.00 B)

Epoch 1/15
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 7s 3ms/step - accuracy: 0.8088 - loss:
0.7304
Epoch 2/15
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 4s 2ms/step - accuracy: 0.9118 - loss:
0.3107
Epoch 3/15
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 4s 2ms/step - accuracy: 0.9201 - loss:
0.2881
Epoch 4/15
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 4s 2ms/step - accuracy: 0.9231 - loss:
0.2704
Epoch 5/15
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 3s 2ms/step - accuracy: 0.9262 - loss:
0.2650
Epoch 6/15
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 4s 2ms/step - accuracy: 0.9287 - loss:
0.2600
Epoch 7/15
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 4s 2ms/step - accuracy: 0.9285 - loss:
0.2563
Epoch 8/15
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 3s 2ms/step - accuracy: 0.9315 - loss:
0.2504
Epoch 9/15
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 4s 2ms/step - accuracy: 0.9302 - loss:
0.2547
Epoch 10/15
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 5s 2ms/step - accuracy: 0.9306 - loss:
0.2523
Epoch 11/15
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 4s 2ms/step - accuracy: 0.9324 - loss:
0.2391
Epoch 12/15
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 3s 2ms/step - accuracy: 0.9327 - loss:
0.2431
Epoch 13/15
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 4s 2ms/step - accuracy: 0.9320 - loss:
0.2450
Epoch 14/15
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 4s 2ms/step - accuracy: 0.9329 - loss:
```

```
0.2422
Epoch 15/15
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 3s 2ms/step - accuracy: 0.9322 - loss:
0.2469
313/313 ━━━━━━━━━━━━━━━━━━━━ 1s 2ms/step - accuracy: 0.9167 - loss:
0.3005
Test accuracy with batch size 32 : 0.9271000027656555

Training with batch size: 64

Model: "sequential_13"

┏━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━┳━━━━━━━━━━━━━━━━━━━━━━━━━━━━━┳━━━━━━━━━━━━━━━━━
━━━┓
┃ Layer (type)                         ┃ Output Shape                ┃
Param # ┃
┡━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━╇━━━━━━━━━━━━━━━━━━━━━━━━━━━━━╇━━━━━━━━━━━━━━━━━
━━━┩
│ dense_13 (Dense)                     │ (None, 10)                  │
7,850 │
└──────────────────────────────────────┴─────────────────────────────┴─────────────────
━━━━┘

 Total params: 7,850 (30.66 KB)

 Trainable params: 7,850 (30.66 KB)

 Non-trainable params: 0 (0.00 B)

Epoch 1/15
938/938 ━━━━━━━━━━━━━━━━━━━━ 3s 2ms/step - accuracy: 0.7833 - loss:
0.8541
Epoch 2/15
938/938 ━━━━━━━━━━━━━━━━━━━━ 3s 3ms/step - accuracy: 0.9094 - loss:
0.3327
Epoch 3/15
938/938 ━━━━━━━━━━━━━━━━━━━━ 2s 2ms/step - accuracy: 0.9193 - loss:
0.2937
Epoch 4/15
938/938 ━━━━━━━━━━━━━━━━━━━━ 2s 2ms/step - accuracy: 0.9206 - loss:
0.2841
Epoch 5/15
938/938 ━━━━━━━━━━━━━━━━━━━━ 2s 2ms/step - accuracy: 0.9221 - loss:
0.2753
Epoch 6/15
938/938 ━━━━━━━━━━━━━━━━━━━━ 2s 2ms/step - accuracy: 0.9267 - loss:
0.2660
Epoch 7/15
938/938 ━━━━━━━━━━━━━━━━━━━━ 3s 3ms/step - accuracy: 0.9266 - loss:
0.2623
Epoch 8/15
```

```
938/938 ━━━━━━━━━━━━━━━━━━━━ 4s 4ms/step - accuracy: 0.9277 - loss:
0.2570
Epoch 9/15
938/938 ━━━━━━━━━━━━━━━━━━━━ 4s 4ms/step - accuracy: 0.9300 - loss:
0.2561
Epoch 10/15
938/938 ━━━━━━━━━━━━━━━━━━━━ 5s 4ms/step - accuracy: 0.9313 - loss:
0.2456
Epoch 11/15
938/938 ━━━━━━━━━━━━━━━━━━━━ 4s 3ms/step - accuracy: 0.9315 - loss:
0.2451
Epoch 12/15
938/938 ━━━━━━━━━━━━━━━━━━━━ 3s 3ms/step - accuracy: 0.9313 - loss:
0.2512
Epoch 13/15
938/938 ━━━━━━━━━━━━━━━━━━━━ 4s 3ms/step - accuracy: 0.9309 - loss:
0.2535
Epoch 14/15
938/938 ━━━━━━━━━━━━━━━━━━━━ 2s 3ms/step - accuracy: 0.9323 - loss:
0.2493
Epoch 15/15
938/938 ━━━━━━━━━━━━━━━━━━━━ 2s 2ms/step - accuracy: 0.9329 - loss:
0.2413
313/313 ━━━━━━━━━━━━━━━━━━━━ 1s 2ms/step - accuracy: 0.9166 - loss:
0.2993
Test accuracy with batch size 64 : 0.927299976348877

Training with batch size: 128

Model: "sequential_14"

┏━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━┳━━━━━━━━━━━━━━━━━━━━━━━━━━━━━┳━━━━━━━━━━━━━━━━━
━━┓
┃ Layer (type)                         ┃ Output Shape                ┃
Param # ┃
┡━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━╇━━━━━━━━━━━━━━━━━━━━━━━━━━━━━╇━━━━━━━━━━━━━━━━━
━━┩
│ dense_14 (Dense)                     │ (None, 10)                  │
7,850 │
└──────────────────────────────────────┴─────────────────────────────┴─────────────────
──┘

 Total params: 7,850 (30.66 KB)

 Trainable params: 7,850 (30.66 KB)

 Non-trainable params: 0 (0.00 B)

Epoch 1/15
469/469 ━━━━━━━━━━━━━━━━━━━━ 2s 3ms/step - accuracy: 0.7391 - loss:
1.0215
```

```
Epoch 2/15
469/469 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - accuracy: 0.8985 - loss:
0.3783
Epoch 3/15
469/469 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - accuracy: 0.9120 - loss:
0.3196
Epoch 4/15
469/469 ━━━━━━━━━━━━━━━━━━━━ 1s 2ms/step - accuracy: 0.9164 - loss:
0.2994
Epoch 5/15
469/469 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - accuracy: 0.9208 - loss:
0.2842
Epoch 6/15
469/469 ━━━━━━━━━━━━━━━━━━━━ 2s 3ms/step - accuracy: 0.9225 - loss:
0.2793
Epoch 7/15
469/469 ━━━━━━━━━━━━━━━━━━━━ 2s 4ms/step - accuracy: 0.9235 - loss:
0.2753
Epoch 8/15
469/469 ━━━━━━━━━━━━━━━━━━━━ 1s 2ms/step - accuracy: 0.9255 - loss:
0.2661
Epoch 9/15
469/469 ━━━━━━━━━━━━━━━━━━━━ 1s 2ms/step - accuracy: 0.9274 - loss:
0.2680
Epoch 10/15
469/469 ━━━━━━━━━━━━━━━━━━━━ 1s 2ms/step - accuracy: 0.9277 - loss:
0.2588
Epoch 11/15
469/469 ━━━━━━━━━━━━━━━━━━━━ 1s 2ms/step - accuracy: 0.9287 - loss:
0.2613
Epoch 12/15
469/469 ━━━━━━━━━━━━━━━━━━━━ 1s 2ms/step - accuracy: 0.9292 - loss:
0.2557
Epoch 13/15
469/469 ━━━━━━━━━━━━━━━━━━━━ 1s 2ms/step - accuracy: 0.9291 - loss:
0.2520
Epoch 14/15
469/469 ━━━━━━━━━━━━━━━━━━━━ 1s 2ms/step - accuracy: 0.9296 - loss:
0.2493
Epoch 15/15
469/469 ━━━━━━━━━━━━━━━━━━━━ 1s 2ms/step - accuracy: 0.9332 - loss:
0.2457
313/313 ━━━━━━━━━━━━━━━━━━━━ 1s 2ms/step - accuracy: 0.9160 - loss:
0.2997
Test accuracy with batch size 128 : 0.9275000095367432

Training with batch size: 256

Model: "sequential_15"
```

```
┏━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━┳━━━━━━━━━━━━━━━━━━━━━━━━┳━━━━━━━━━━━━━━━┓
┃ Layer (type)                    ┃ Output Shape           ┃       Param # ┃
┡━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━╇━━━━━━━━━━━━━━━━━━━━━━━━╇━━━━━━━━━━━━━━━┩
│ dense_15 (Dense)                │ (None, 10)             │         7,850 │
└─────────────────────────────────┴────────────────────────┴───────────────┘

 Total params: 7,850 (30.66 KB)

 Trainable params: 7,850 (30.66 KB)

 Non-trainable params: 0 (0.00 B)

Epoch 1/15
235/235 ━━━━━━━━━━━━━━━━━━━━ 2s 5ms/step - accuracy: 0.6377 - loss:
1.2971
Epoch 2/15
235/235 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - accuracy: 0.8837 - loss:
0.4535
Epoch 3/15
235/235 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - accuracy: 0.8992 - loss:
0.3755
Epoch 4/15
235/235 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - accuracy: 0.9098 - loss:
0.3316
Epoch 5/15
235/235 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - accuracy: 0.9129 - loss:
0.3177
Epoch 6/15
235/235 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - accuracy: 0.9171 - loss:
0.3057
Epoch 7/15
235/235 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - accuracy: 0.9196 - loss:
0.2886
Epoch 8/15
235/235 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - accuracy: 0.9242 - loss:
0.2769
Epoch 9/15
235/235 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - accuracy: 0.9214 - loss:
0.2820
Epoch 10/15
235/235 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - accuracy: 0.9228 - loss:
0.2760
Epoch 11/15
235/235 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - accuracy: 0.9251 - loss:
0.2666
```

```
Epoch 12/15
235/235 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - accuracy: 0.9243 - loss:
0.2694
Epoch 13/15
235/235 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - accuracy: 0.9269 - loss:
0.2626
Epoch 14/15
235/235 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - accuracy: 0.9295 - loss:
0.2591
Epoch 15/15
235/235 ━━━━━━━━━━━━━━━━━━━━ 1s 5ms/step - accuracy: 0.9277 - loss:
0.2611
313/313 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - accuracy: 0.9154 - loss:
0.3028
Test accuracy with batch size 256 : 0.9259999990463257
```

Change the number of epochs

```python
import tensorflow as tf
from tensorflow.keras.datasets import mnist
from tensorflow.keras.utils import to_categorical
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense, Flatten
import numpy as np

# Load dataset
(X_train, y_train), (X_test, y_test) = mnist.load_data()

# Normalize
X_train = X_train / 255.0
X_test = X_test / 255.0

# Flatten
X_train = X_train.reshape(-1, 784)
X_test = X_test.reshape(-1, 784)

# One-hot encoding
y_train = to_categorical(y_train, 10)
y_test = to_categorical(y_test, 10)

# Different epoch values
epochs_list = [10, 15, 20]

for ep in epochs_list:

    print("\nTraining with epochs:", ep)

    # Model
    model = Sequential([
        Dense(10, activation='softmax', input_shape=(784,))
```

```
    ])

    model.compile(optimizer='adam',
                  loss='categorical_crossentropy',
                  metrics=['accuracy'])

    model.summary()

    # Train model
    model.fit(X_train, y_train, epochs=ep, batch_size=64)

    # Evaluate
    test_loss, test_acc = model.evaluate(X_test, y_test)

    print("Epochs:", ep, "| Test accuracy:", test_acc)


Training with epochs: 10

Model: "sequential_30"

┏━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━┳━━━━━━━━━━━━━━━━━━━━━━━━━━━━━┳━━━━━━━━━━━━━━━━━
┓
┃ Layer (type)                         ┃ Output Shape                ┃
Param # ┃
┡━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━╇━━━━━━━━━━━━━━━━━━━━━━━━━━━━━╇━━━━━━━━━━━━━━━━━
┩
│ dense_30 (Dense)                     │ (None, 10)                  │
7,850 │
└──────────────────────────────────────┴─────────────────────────────┴─────────────────
┘

 Total params: 7,850 (30.66 KB)

 Trainable params: 7,850 (30.66 KB)

 Non-trainable params: 0 (0.00 B)

Epoch 1/10
938/938 ━━━━━━━━━━━━━━━━━━━━ 4s 4ms/step - accuracy: 0.7850 - loss:
0.8524
Epoch 2/10
938/938 ━━━━━━━━━━━━━━━━━━━━ 4s 2ms/step - accuracy: 0.9090 - loss:
0.3332
Epoch 3/10
938/938 ━━━━━━━━━━━━━━━━━━━━ 3s 3ms/step - accuracy: 0.9157 - loss:
0.3022
Epoch 4/10
938/938 ━━━━━━━━━━━━━━━━━━━━ 2s 2ms/step - accuracy: 0.9216 - loss:
0.2814
Epoch 5/10
```

```
938/938 ━━━━━━━━━━━━━━━━━━━━ 2s 2ms/step - accuracy: 0.9236 - loss:
0.2753
Epoch 6/10
938/938 ━━━━━━━━━━━━━━━━━━━━ 2s 2ms/step - accuracy: 0.9269 - loss:
0.2673
Epoch 7/10
938/938 ━━━━━━━━━━━━━━━━━━━━ 2s 2ms/step - accuracy: 0.9271 - loss:
0.2605
Epoch 8/10
938/938 ━━━━━━━━━━━━━━━━━━━━ 2s 2ms/step - accuracy: 0.9283 - loss:
0.2587
Epoch 9/10
938/938 ━━━━━━━━━━━━━━━━━━━━ 3s 3ms/step - accuracy: 0.9290 - loss:
0.2551
Epoch 10/10
938/938 ━━━━━━━━━━━━━━━━━━━━ 4s 2ms/step - accuracy: 0.9298 - loss:
0.2538
313/313 ━━━━━━━━━━━━━━━━━━━━ 1s 2ms/step - accuracy: 0.9147 - loss:
0.3030
Epochs: 10 | Test accuracy: 0.9258999824523926

Training with epochs: 15

Model: "sequential_31"

┏━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━┳━━━━━━━━━━━━━━━━━━━━━━━━━━━━━┳━━━━━━━━━━━━━━━━━
━━━━┓
┃ Layer (type)                         ┃ Output Shape                ┃
Param # ┃
┡━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━╇━━━━━━━━━━━━━━━━━━━━━━━━━━━━━╇━━━━━━━━━━━━━━━━━
━━━━┩
│ dense_31 (Dense)                     │ (None, 10)                  │
7,850 │
└──────────────────────────────────────┴─────────────────────────────┴─────────────────
────┘

 Total params: 7,850 (30.66 KB)

 Trainable params: 7,850 (30.66 KB)

 Non-trainable params: 0 (0.00 B)

Epoch 1/15
938/938 ━━━━━━━━━━━━━━━━━━━━ 2s 2ms/step - accuracy: 0.7792 - loss:
0.8587
Epoch 2/15
938/938 ━━━━━━━━━━━━━━━━━━━━ 2s 2ms/step - accuracy: 0.9078 - loss:
0.3371
Epoch 3/15
938/938 ━━━━━━━━━━━━━━━━━━━━ 3s 3ms/step - accuracy: 0.9151 - loss:
0.3043
```

```
Epoch 4/15
938/938 ━━━━━━━━━━━━━━━━━━━━ 4s 2ms/step - accuracy: 0.9201 - loss:
0.2829
Epoch 5/15
938/938 ━━━━━━━━━━━━━━━━━━━━ 2s 2ms/step - accuracy: 0.9223 - loss:
0.2750
Epoch 6/15
938/938 ━━━━━━━━━━━━━━━━━━━━ 2s 2ms/step - accuracy: 0.9257 - loss:
0.2676
Epoch 7/15
938/938 ━━━━━━━━━━━━━━━━━━━━ 2s 2ms/step - accuracy: 0.9270 - loss:
0.2629
Epoch 8/15
938/938 ━━━━━━━━━━━━━━━━━━━━ 3s 3ms/step - accuracy: 0.9280 - loss:
0.2606
Epoch 9/15
938/938 ━━━━━━━━━━━━━━━━━━━━ 2s 2ms/step - accuracy: 0.9284 - loss:
0.2577
Epoch 10/15
938/938 ━━━━━━━━━━━━━━━━━━━━ 2s 2ms/step - accuracy: 0.9309 - loss:
0.2523
Epoch 11/15
938/938 ━━━━━━━━━━━━━━━━━━━━ 2s 2ms/step - accuracy: 0.9302 - loss:
0.2521
Epoch 12/15
938/938 ━━━━━━━━━━━━━━━━━━━━ 2s 2ms/step - accuracy: 0.9311 - loss:
0.2464
Epoch 13/15
938/938 ━━━━━━━━━━━━━━━━━━━━ 2s 2ms/step - accuracy: 0.9309 - loss:
0.2457
Epoch 14/15
938/938 ━━━━━━━━━━━━━━━━━━━━ 3s 3ms/step - accuracy: 0.9339 - loss:
0.2457
Epoch 15/15
938/938 ━━━━━━━━━━━━━━━━━━━━ 2s 2ms/step - accuracy: 0.9333 - loss:
0.2439
313/313 ━━━━━━━━━━━━━━━━━━━━ 1s 2ms/step - accuracy: 0.9169 - loss:
0.2978
Epochs: 15 | Test accuracy: 0.9269999861717224

Training with epochs: 20

Model: "sequential_32"

┏━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━┳━━━━━━━━━━━━━━━━━━━━━━━━━━━━━┳━━━━━━━━━━━━━━━━━
━━┓
┃ Layer (type)                         ┃ Output Shape                ┃
Param # ┃
┡━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━╇━━━━━━━━━━━━━━━━━━━━━━━━━━━━━╇━━━━━━━━━━━━━━━━━
━━┩
```

```
| dense_32 (Dense)                 | (None, 10)              |
7,850 |
└──────────────────────────────────┴─────────────────────────┴──────────
──────┘

 Total params: 7,850 (30.66 KB)

 Trainable params: 7,850 (30.66 KB)

 Non-trainable params: 0 (0.00 B)

Epoch 1/20
938/938 ━━━━━━━━━━━━━━━━━━━━ 2s 2ms/step - accuracy: 0.7607 - loss:
0.8978
Epoch 2/20
938/938 ━━━━━━━━━━━━━━━━━━━━ 2s 2ms/step - accuracy: 0.9070 - loss:
0.3359
Epoch 3/20
938/938 ━━━━━━━━━━━━━━━━━━━━ 2s 2ms/step - accuracy: 0.9167 - loss:
0.3020
Epoch 4/20
938/938 ━━━━━━━━━━━━━━━━━━━━ 3s 3ms/step - accuracy: 0.9226 - loss:
0.2795
Epoch 5/20
938/938 ━━━━━━━━━━━━━━━━━━━━ 5s 3ms/step - accuracy: 0.9236 - loss:
0.2766
Epoch 6/20
938/938 ━━━━━━━━━━━━━━━━━━━━ 5s 2ms/step - accuracy: 0.9261 - loss:
0.2666
Epoch 7/20
938/938 ━━━━━━━━━━━━━━━━━━━━ 3s 3ms/step - accuracy: 0.9261 - loss:
0.2650
Epoch 8/20
938/938 ━━━━━━━━━━━━━━━━━━━━ 3s 3ms/step - accuracy: 0.9275 - loss:
0.2610
Epoch 9/20
938/938 ━━━━━━━━━━━━━━━━━━━━ 4s 4ms/step - accuracy: 0.9284 - loss:
0.2624
Epoch 10/20
938/938 ━━━━━━━━━━━━━━━━━━━━ 5s 5ms/step - accuracy: 0.9293 - loss:
0.2523
Epoch 11/20
938/938 ━━━━━━━━━━━━━━━━━━━━ 5s 6ms/step - accuracy: 0.9301 - loss:
0.2544
Epoch 12/20
938/938 ━━━━━━━━━━━━━━━━━━━━ 7s 2ms/step - accuracy: 0.9314 - loss:
0.2493
Epoch 13/20
938/938 ━━━━━━━━━━━━━━━━━━━━ 2s 2ms/step - accuracy: 0.9333 - loss:
0.2445
```

```
Epoch 14/20
938/938 ━━━━━━━━━━━━━━━━━━━━ 3s 3ms/step - accuracy: 0.9294 - loss:
0.2522
Epoch 15/20
938/938 ━━━━━━━━━━━━━━━━━━━━ 2s 2ms/step - accuracy: 0.9329 - loss:
0.2411
Epoch 16/20
938/938 ━━━━━━━━━━━━━━━━━━━━ 2s 2ms/step - accuracy: 0.9322 - loss:
0.2438
Epoch 17/20
938/938 ━━━━━━━━━━━━━━━━━━━━ 2s 2ms/step - accuracy: 0.9346 - loss:
0.2368
Epoch 18/20
938/938 ━━━━━━━━━━━━━━━━━━━━ 2s 2ms/step - accuracy: 0.9318 - loss:
0.2456
Epoch 19/20
938/938 ━━━━━━━━━━━━━━━━━━━━ 3s 2ms/step - accuracy: 0.9340 - loss:
0.2398
Epoch 20/20
938/938 ━━━━━━━━━━━━━━━━━━━━ 3s 3ms/step - accuracy: 0.9359 - loss:
0.2352
313/313 ━━━━━━━━━━━━━━━━━━━━ 1s 2ms/step - accuracy: 0.9177 - loss:
0.2992
Epochs: 20 | Test accuracy: 0.9283000230789185
```

Accuracy and Loss curves

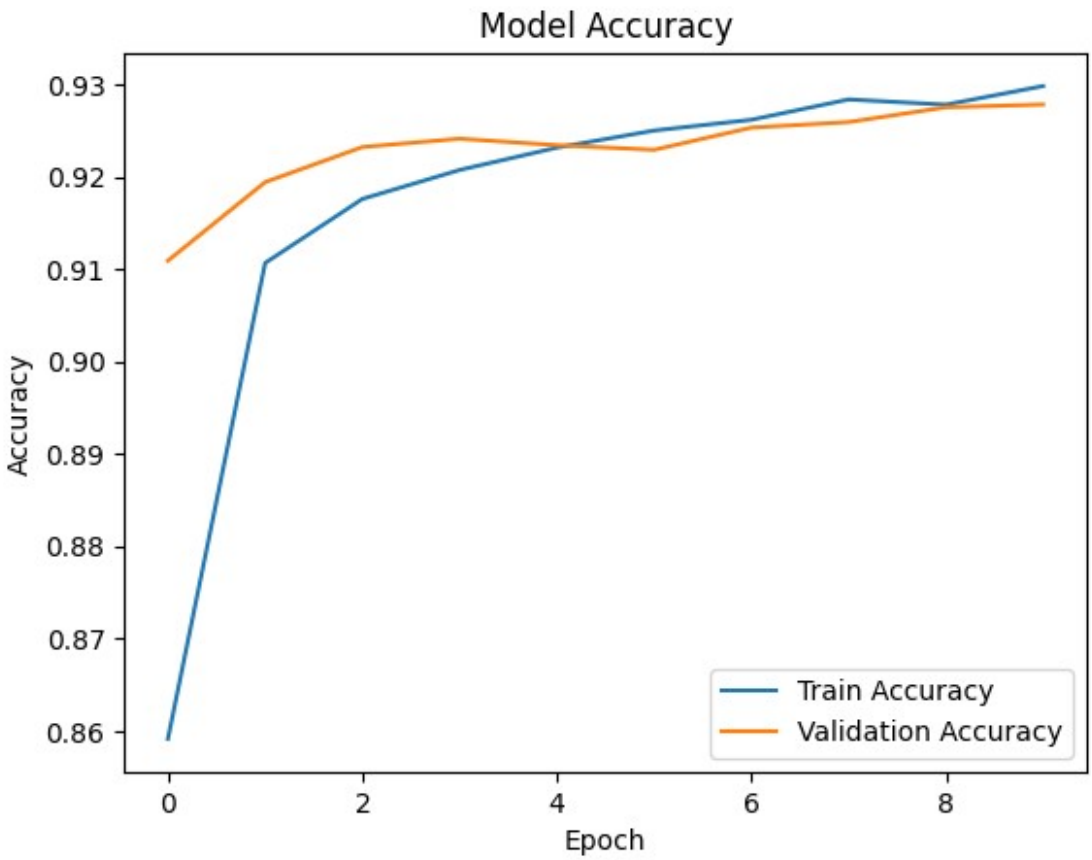
```python
import tensorflow as tf
from tensorflow.keras.datasets import mnist
from tensorflow.keras.utils import to_categorical
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense
import numpy as np
import matplotlib.pyplot as plt

# Load dataset
(X_train, y_train), (X_test, y_test) = mnist.load_data()

# Normalize
X_train = X_train / 255.0
X_test = X_test / 255.0

# Flatten
X_train = X_train.reshape(-1, 784)
X_test = X_test.reshape(-1, 784)

# One-hot encoding
y_train = to_categorical(y_train, 10)
y_test = to_categorical(y_test, 10)
```

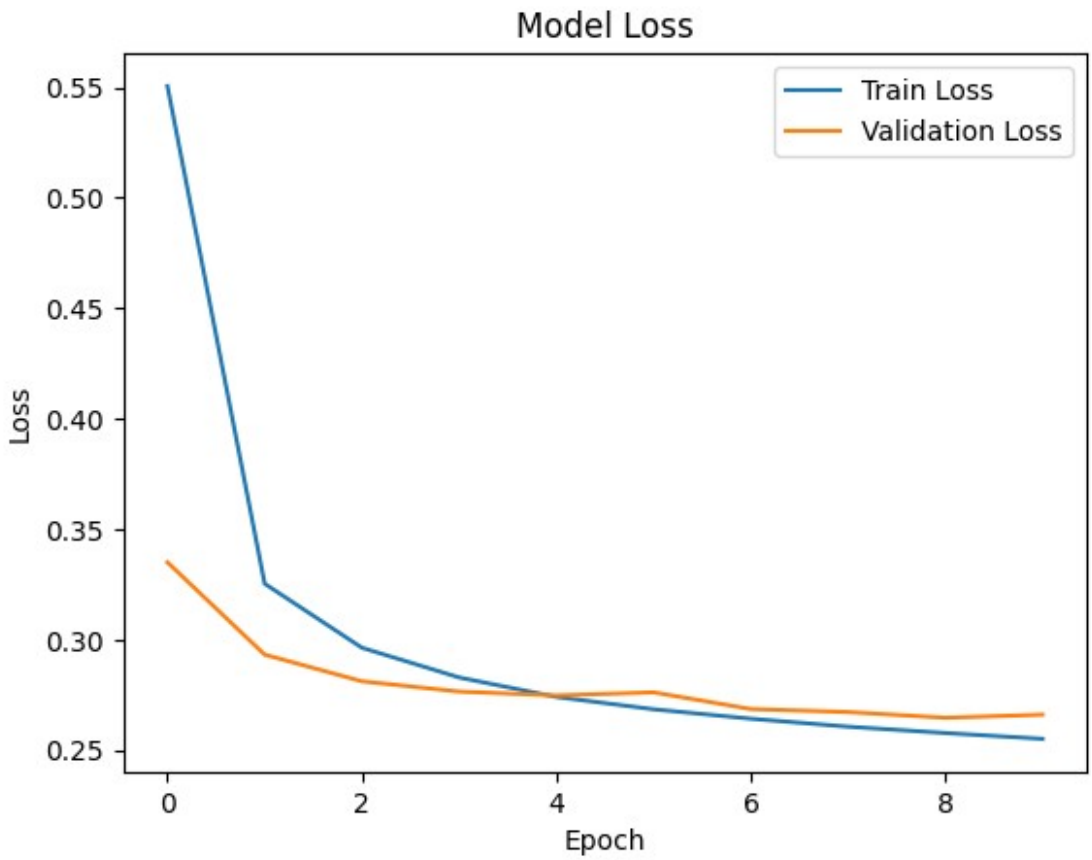
```python
# Model
model = Sequential([
    Dense(10, activation='softmax', input_shape=(784,))
])

model.compile(optimizer='adam',
              loss='categorical_crossentropy',
              metrics=['accuracy'])

model.summary()

# Train model and store history
history = model.fit(
    X_train, y_train,
    epochs=10,
    batch_size=64,
    validation_data=(X_test, y_test)
)

# Evaluate model
test_loss, test_acc = model.evaluate(X_test, y_test)
print("Test accuracy:", test_acc)

# Plot Accuracy Curve
plt.figure()
plt.plot(history.history['accuracy'], label='Train Accuracy')
plt.plot(history.history['val_accuracy'], label='Validation Accuracy')
plt.title('Model Accuracy')
plt.xlabel('Epoch')
plt.ylabel('Accuracy')
plt.legend()
plt.show()

# Plot Loss Curve
plt.figure()
plt.plot(history.history['loss'], label='Train Loss')
plt.plot(history.history['val_loss'], label='Validation Loss')
plt.title('Model Loss')
plt.xlabel('Epoch')
plt.ylabel('Loss')
plt.legend()
plt.show()
```

```
Model: "sequential_33"
```

```
┏━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━┳━━━━━━━━━━━━━━━━━━━━━━━━━━━━━┳━━━━━━━━━━━━━━━━━┓
┃ Layer (type)                         ┃ Output Shape                ┃
Param # ┃
```

```
┣━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━╋━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━╋━━━━━━━━━━━━━━━━
━━━━━━━━┫
│ dense_33 (Dense)                                     │ (None, 10)                             │
7,850 │
└──────────────────────────────────────────────────────┴────────────────────────────────────────┴────────────────
────────┘

 Total params: 7,850 (30.66 KB)

 Trainable params: 7,850 (30.66 KB)

 Non-trainable params: 0 (0.00 B)

Epoch 1/10
938/938 ━━━━━━━━━━━━━━━━━━━━ 3s 2ms/step - accuracy: 0.7769 - loss:
0.8626 - val_accuracy: 0.9109 - val_loss: 0.3350
Epoch 2/10
938/938 ━━━━━━━━━━━━━━━━━━━━ 2s 2ms/step - accuracy: 0.9087 - loss:
0.3328 - val_accuracy: 0.9194 - val_loss: 0.2932
Epoch 3/10
938/938 ━━━━━━━━━━━━━━━━━━━━ 3s 3ms/step - accuracy: 0.9157 - loss:
0.2998 - val_accuracy: 0.9232 - val_loss: 0.2811
Epoch 4/10
938/938 ━━━━━━━━━━━━━━━━━━━━ 3s 3ms/step - accuracy: 0.9198 - loss:
0.2865 - val_accuracy: 0.9241 - val_loss: 0.2764
Epoch 5/10
938/938 ━━━━━━━━━━━━━━━━━━━━ 2s 2ms/step - accuracy: 0.9237 - loss:
0.2713 - val_accuracy: 0.9234 - val_loss: 0.2747
Epoch 6/10
938/938 ━━━━━━━━━━━━━━━━━━━━ 2s 2ms/step - accuracy: 0.9252 - loss:
0.2652 - val_accuracy: 0.9229 - val_loss: 0.2761
Epoch 7/10
938/938 ━━━━━━━━━━━━━━━━━━━━ 2s 2ms/step - accuracy: 0.9280 - loss:
0.2564 - val_accuracy: 0.9253 - val_loss: 0.2685
Epoch 8/10
938/938 ━━━━━━━━━━━━━━━━━━━━ 2s 2ms/step - accuracy: 0.9297 - loss:
0.2580 - val_accuracy: 0.9259 - val_loss: 0.2672
Epoch 9/10
938/938 ━━━━━━━━━━━━━━━━━━━━ 3s 3ms/step - accuracy: 0.9287 - loss:
0.2583 - val_accuracy: 0.9275 - val_loss: 0.2646
Epoch 10/10
938/938 ━━━━━━━━━━━━━━━━━━━━ 2s 2ms/step - accuracy: 0.9274 - loss:
0.2626 - val_accuracy: 0.9278 - val_loss: 0.2660
313/313 ━━━━━━━━━━━━━━━━━━━━ 1s 2ms/step - accuracy: 0.9164 - loss:
0.2998
Test accuracy: 0.9277999997138977
```


Model Accuracy
0.93
0.92
0.91
0.90
0.89
0.88
0.87
0.86
Accuracy
0
2
4
6
8
Epoch
Train Accuracy
Validation Accuracy



Add next layer using reloc

hidden layer(32,ReLu)

```
import tensorflow as tf
from tensorflow.keras.datasets import mnist
from tensorflow.keras.utils import to_categorical
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense
import matplotlib.pyplot as plt
import numpy as np

# 1. Load MNIST dataset
(X_train, y_train), (X_test, y_test) = mnist.load_data()

# 2. Normalize pixel values
X_train = X_train / 255.0
X_test = X_test / 255.0

# 3. Flatten images (28x28 → 784)
X_train = X_train.reshape(-1, 784)
```

```
X_test = X_test.reshape(-1, 784)

# 4. One-hot encoding
y_train = to_categorical(y_train, 10)
y_test = to_categorical(y_test, 10)

# 5. Build model (Added hidden layer with 32 neurons)
model = Sequential([
    Dense(32, activation='relu', input_shape=(784,)),   # Hidden layer
    Dense(10, activation='softmax')                     # Output layer
])

# 6. Compile model
model.compile(
    optimizer='adam',
    loss='categorical_crossentropy',
    metrics=['accuracy']
)

# 7. Model summary
model.summary()

# 8. Train model
history = model.fit(
    X_train,
    y_train,
    epochs=10,
    batch_size=64,
    validation_data=(X_test, y_test)
)

# 9. Evaluate model
test_loss, test_acc = model.evaluate(X_test, y_test)
print("Test accuracy:", test_acc)

# 10. Plot Accuracy Curve
plt.figure()
plt.plot(history.history['accuracy'])
plt.plot(history.history['val_accuracy'])
plt.title("Model Accuracy")
plt.xlabel("Epoch")
plt.ylabel("Accuracy")
plt.legend(["Train", "Validation"])
plt.show()

# 11. Plot Loss Curve
plt.figure()
plt.plot(history.history['loss'])
plt.plot(history.history['val_loss'])
plt.title("Model Loss")
```

```
plt.xlabel("Epoch")
plt.ylabel("Loss")
plt.legend(["Train", "Validation"])
plt.show()
```

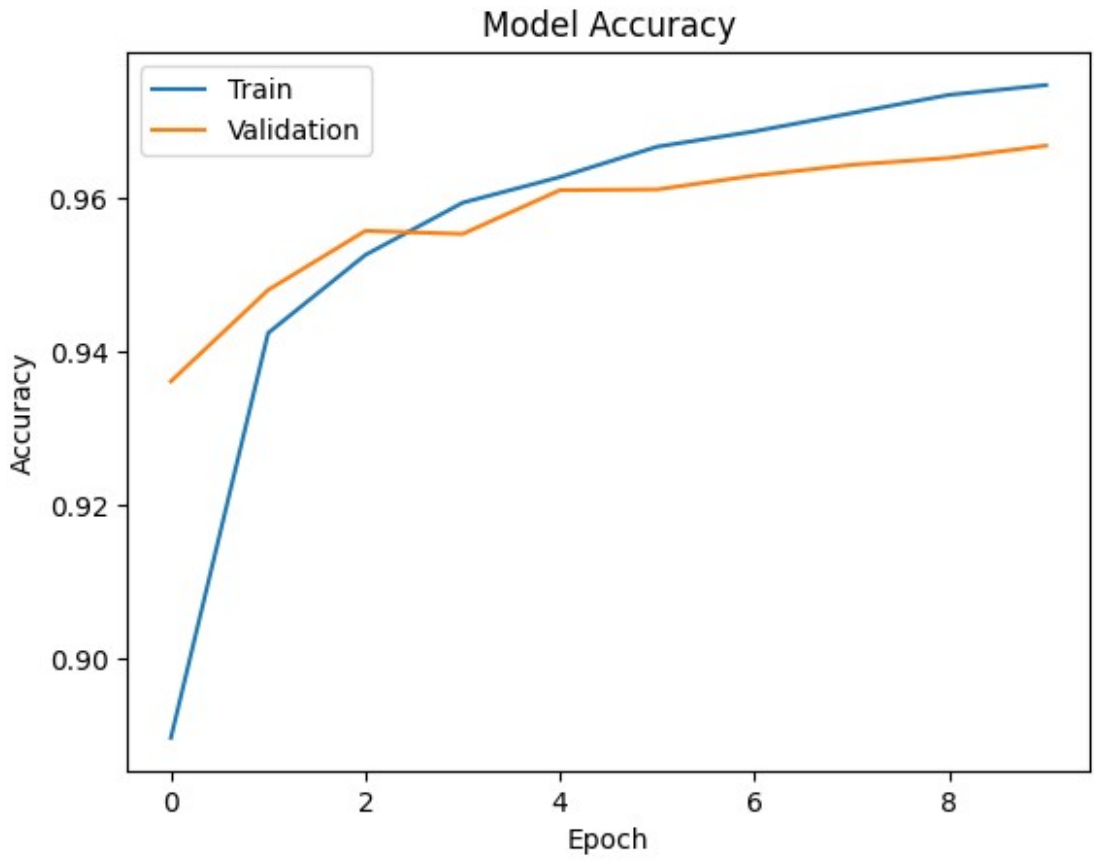
```
Model: "sequential_35"
```

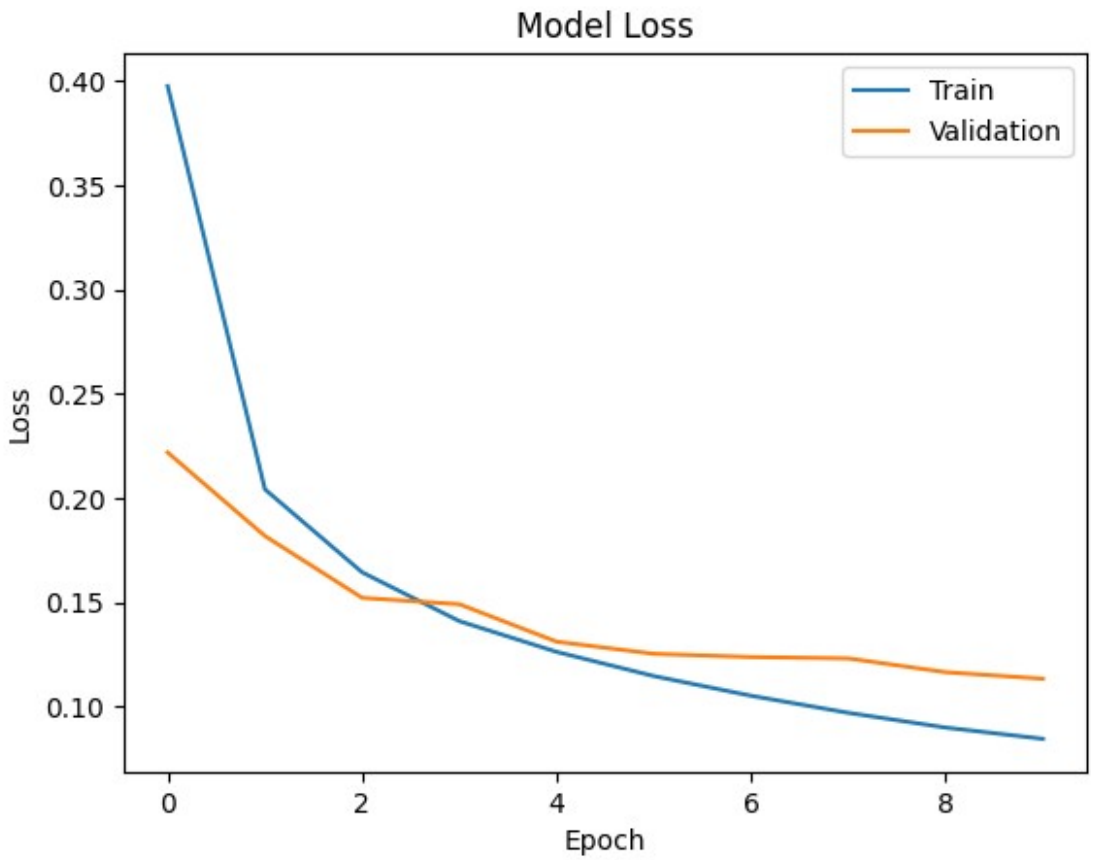
```
┏━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━┳━━━━━━━━━━━━━━━━━━━━━━━━┳━━━━━━━━━━━━━━━┓
┃ Layer (type)                    ┃ Output Shape           ┃       Param # ┃
┡━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━╇━━━━━━━━━━━━━━━━━━━━━━━━╇━━━━━━━━━━━━━━━┩
│ dense_35 (Dense)                │ (None, 32)             │        25,120 │
├─────────────────────────────────┼────────────────────────┼───────────────┤
│ dense_36 (Dense)                │ (None, 10)             │           330 │
└─────────────────────────────────┴────────────────────────┴───────────────┘
```

```
 Total params: 25,450 (99.41 KB)

 Trainable params: 25,450 (99.41 KB)

 Non-trainable params: 0 (0.00 B)
```

```
Epoch 1/10
938/938 ━━━━━━━━━━━━━━━━━━━━ 6s 5ms/step - accuracy: 0.8079 - loss:
0.6729 - val_accuracy: 0.9361 - val_loss: 0.2218
Epoch 2/10
938/938 ━━━━━━━━━━━━━━━━━━━━ 3s 3ms/step - accuracy: 0.9414 - loss:
0.2103 - val_accuracy: 0.9480 - val_loss: 0.1817
Epoch 3/10
938/938 ━━━━━━━━━━━━━━━━━━━━ 4s 4ms/step - accuracy: 0.9525 - loss:
0.1650 - val_accuracy: 0.9557 - val_loss: 0.1520
Epoch 4/10
938/938 ━━━━━━━━━━━━━━━━━━━━ 3s 3ms/step - accuracy: 0.9592 - loss:
0.1397 - val_accuracy: 0.9553 - val_loss: 0.1490
Epoch 5/10
938/938 ━━━━━━━━━━━━━━━━━━━━ 3s 3ms/step - accuracy: 0.9625 - loss:
0.1250 - val_accuracy: 0.9610 - val_loss: 0.1310
Epoch 6/10
938/938 ━━━━━━━━━━━━━━━━━━━━ 3s 3ms/step - accuracy: 0.9683 - loss:
0.1077 - val_accuracy: 0.9611 - val_loss: 0.1252
Epoch 7/10
938/938 ━━━━━━━━━━━━━━━━━━━━ 3s 3ms/step - accuracy: 0.9695 - loss:
0.1033 - val_accuracy: 0.9629 - val_loss: 0.1236
Epoch 8/10
938/938 ━━━━━━━━━━━━━━━━━━━━ 3s 3ms/step - accuracy: 0.9712 - loss:
```

```
0.0967 - val_accuracy: 0.9643 - val_loss: 0.1230
Epoch 9/10
938/938 ━━━━━━━━━━━━━━━━━━━━ 2s 3ms/step - accuracy: 0.9744 - loss:
0.0893 - val_accuracy: 0.9652 - val_loss: 0.1163
Epoch 10/10
938/938 ━━━━━━━━━━━━━━━━━━━━ 3s 3ms/step - accuracy: 0.9748 - loss:
0.0840 - val_accuracy: 0.9668 - val_loss: 0.1132
313/313 ━━━━━━━━━━━━━━━━━━━━ 1s 2ms/step - accuracy: 0.9629 - loss:
0.1283
Test accuracy: 0.9667999744415283
```





Hidden Layer (64->32)

```
import tensorflow as tf
from tensorflow.keras.datasets import mnist
from tensorflow.keras.utils import to_categorical
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense
import matplotlib.pyplot as plt
import numpy as np

# 1. Load MNIST dataset
(X_train, y_train), (X_test, y_test) = mnist.load_data()

# 2. Normalize pixel values
X_train = X_train / 255.0
X_test = X_test / 255.0

# 3. Flatten images (28x28 → 784)
X_train = X_train.reshape(-1, 784)
X_test = X_test.reshape(-1, 784)

# 4. One-hot encoding
y_train = to_categorical(y_train, 10)
```

```
y_test = to_categorical(y_test, 10)

# 5. Build model (Added 64 and 32 neuron layers)
model = Sequential([
    Dense(64, activation='relu', input_shape=(784,)),  # First hidden
layer
    Dense(32, activation='relu'),                      # Second hidden
layer
    Dense(10, activation='softmax')                    # Output layer
])

# 6. Compile model
model.compile(
    optimizer='adam',
    loss='categorical_crossentropy',
    metrics=['accuracy']
)

# 7. Model summary
model.summary()

# 8. Train model
history = model.fit(
    X_train,
    y_train,
    epochs=10,
    batch_size=64,
    validation_data=(X_test, y_test)
)

# 9. Evaluate model
test_loss, test_acc = model.evaluate(X_test, y_test)
print("Test accuracy:", test_acc)

# 10. Plot Accuracy Curve
plt.figure()
plt.plot(history.history['accuracy'])
plt.plot(history.history['val_accuracy'])
plt.title("Model Accuracy")
plt.xlabel("Epoch")
plt.ylabel("Accuracy")
plt.legend(["Train", "Validation"])
plt.show()

# 11. Plot Loss Curve
plt.figure()
plt.plot(history.history['loss'])
plt.plot(history.history['val_loss'])
plt.title("Model Loss")
plt.xlabel("Epoch")
```

```
plt.ylabel("Loss")
plt.legend(["Train", "Validation"])
plt.show()
```

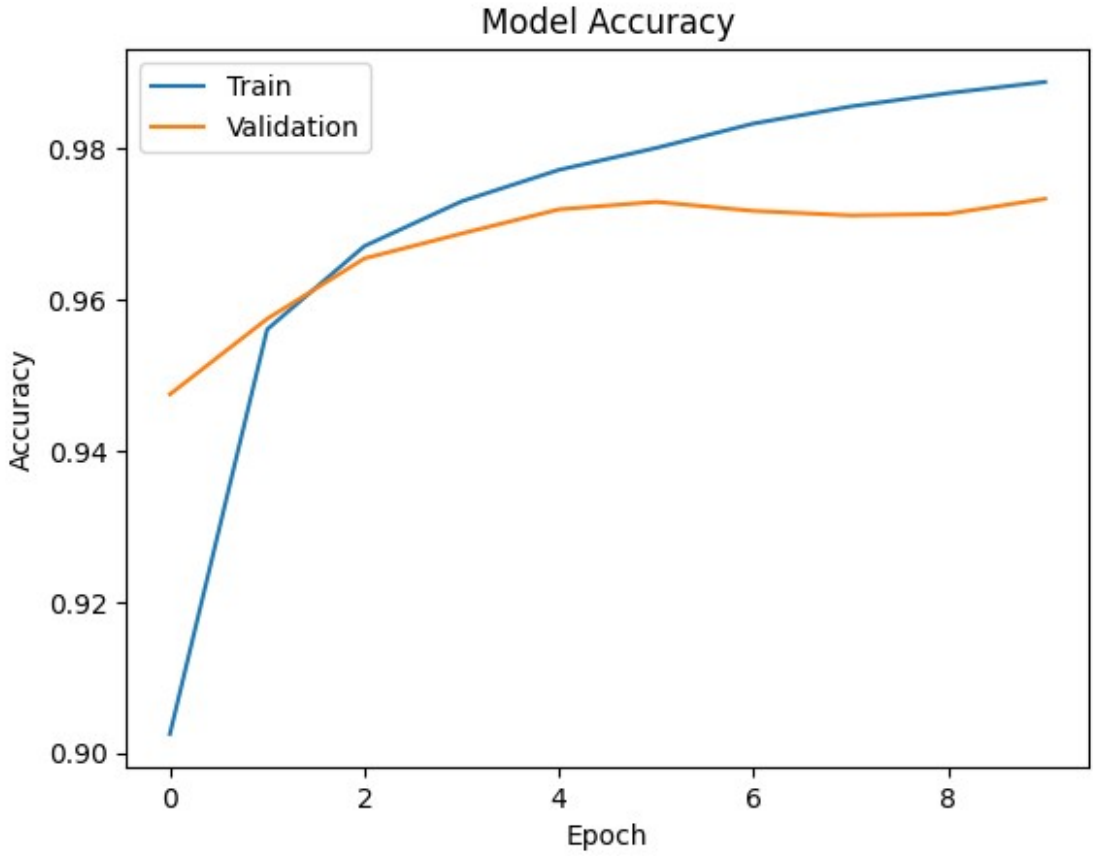
```
Model: "sequential_36"
```

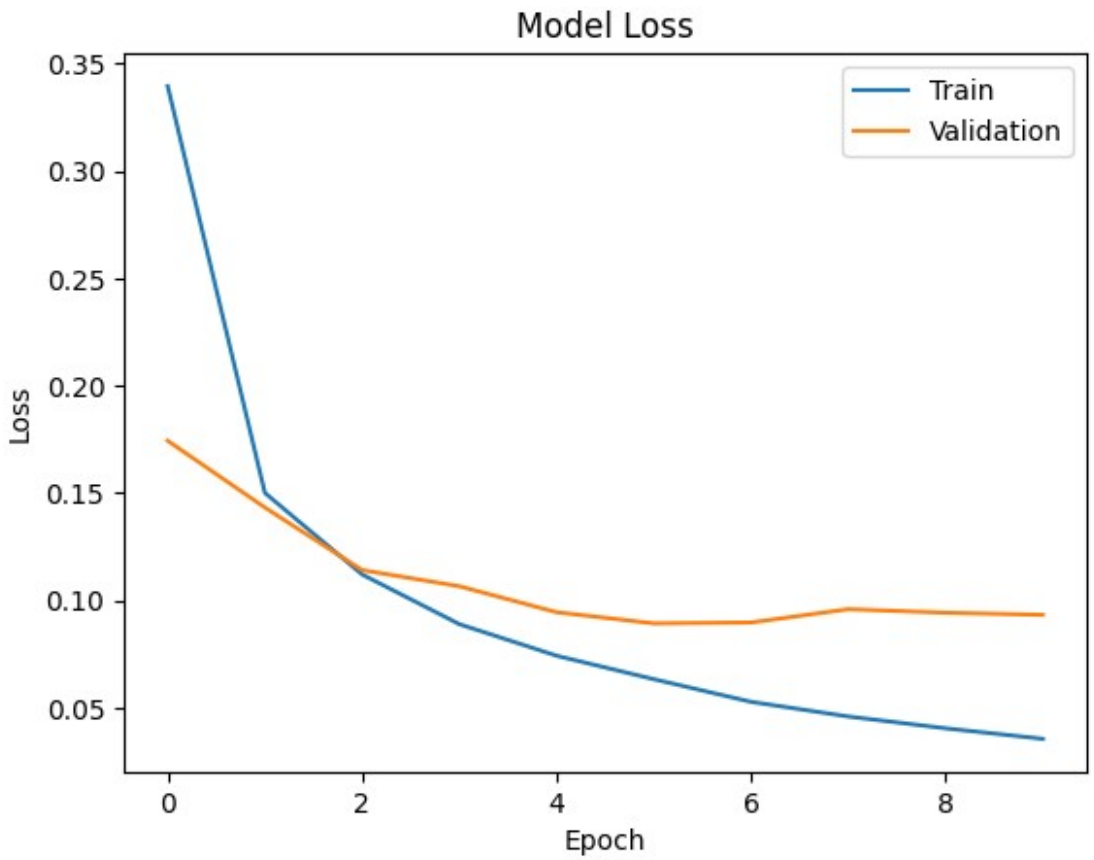
```
┏━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━┳━━━━━━━━━━━━━━━━━━━━━━━━━━━━━┳━━━━━━━━━━━━━━━━━┓
┃ Layer (type)                         ┃ Output Shape                ┃         Param # ┃
┡━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━╇━━━━━━━━━━━━━━━━━━━━━━━━━━━━━╇━━━━━━━━━━━━━━━━━┩
│ dense_37 (Dense)                     │ (None, 64)                  │          50,240 │
├──────────────────────────────────────┼─────────────────────────────┼─────────────────┤
│ dense_38 (Dense)                     │ (None, 32)                  │           2,080 │
├──────────────────────────────────────┼─────────────────────────────┼─────────────────┤
│ dense_39 (Dense)                     │ (None, 10)                  │             330 │
└──────────────────────────────────────┴─────────────────────────────┴─────────────────┘
```

```
 Total params: 52,650 (205.66 KB)

 Trainable params: 52,650 (205.66 KB)

 Non-trainable params: 0 (0.00 B)
```

```
Epoch 1/10
938/938 ━━━━━━━━━━━━━━━━━━━━ 7s 5ms/step - accuracy: 0.8199 - loss:
0.6128 - val_accuracy: 0.9475 - val_loss: 0.1744
Epoch 2/10
938/938 ━━━━━━━━━━━━━━━━━━━━ 4s 4ms/step - accuracy: 0.9528 - loss:
0.1587 - val_accuracy: 0.9575 - val_loss: 0.1433
Epoch 3/10
938/938 ━━━━━━━━━━━━━━━━━━━━ 3s 3ms/step - accuracy: 0.9664 - loss:
0.1156 - val_accuracy: 0.9655 - val_loss: 0.1141
Epoch 4/10
938/938 ━━━━━━━━━━━━━━━━━━━━ 5s 6ms/step - accuracy: 0.9742 - loss:
0.0859 - val_accuracy: 0.9688 - val_loss: 0.1066
Epoch 5/10
938/938 ━━━━━━━━━━━━━━━━━━━━ 7s 7ms/step - accuracy: 0.9780 - loss:
0.0718 - val_accuracy: 0.9720 - val_loss: 0.0945
Epoch 6/10
938/938 ━━━━━━━━━━━━━━━━━━━━ 7s 3ms/step - accuracy: 0.9796 - loss:
0.0645 - val_accuracy: 0.9730 - val_loss: 0.0893
Epoch 7/10
938/938 ━━━━━━━━━━━━━━━━━━━━ 3s 4ms/step - accuracy: 0.9834 - loss:
```

```
0.0521 - val_accuracy: 0.9718 - val_loss: 0.0897
Epoch 8/10
938/938 ━━━━━━━━━━━━━━━━━━━━ 4s 4ms/step - accuracy: 0.9858 - loss:
0.0445 - val_accuracy: 0.9712 - val_loss: 0.0959
Epoch 9/10
938/938 ━━━━━━━━━━━━━━━━━━━━ 3s 3ms/step - accuracy: 0.9886 - loss:
0.0381 - val_accuracy: 0.9714 - val_loss: 0.0943
Epoch 10/10
938/938 ━━━━━━━━━━━━━━━━━━━━ 3s 3ms/step - accuracy: 0.9898 - loss:
0.0324 - val_accuracy: 0.9734 - val_loss: 0.0933
313/313 ━━━━━━━━━━━━━━━━━━━━ 1s 2ms/step - accuracy: 0.9696 - loss:
0.1084
Test accuracy: 0.9733999967575073
```





Hidden layer (128->64->32)

```
import tensorflow as tf
from tensorflow.keras.datasets import mnist
from tensorflow.keras.utils import to_categorical
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense
import matplotlib.pyplot as plt
import numpy as np

# 1. Load MNIST dataset
(X_train, y_train), (X_test, y_test) = mnist.load_data()

# 2. Normalize pixel values
X_train = X_train / 255.0
X_test = X_test / 255.0

# 3. Flatten images (28x28 → 784)
X_train = X_train.reshape(-1, 784)
X_test = X_test.reshape(-1, 784)

# 4. One-hot encoding
y_train = to_categorical(y_train, 10)
```

```
y_test = to_categorical(y_test, 10)

# 5. Build model (Added 128, 64, 32 layers)
model = Sequential([
    Dense(128, activation='relu', input_shape=(784,)),  # Hidden layer
1
    Dense(64, activation='relu'),                       # Hidden layer
2
    Dense(32, activation='relu'),                       # Hidden layer
3
    Dense(10, activation='softmax')                     # Output layer
])

# 6. Compile model
model.compile(
    optimizer='adam',
    loss='categorical_crossentropy',
    metrics=['accuracy']
)

# 7. Model summary
model.summary()

# 8. Train model
history = model.fit(
    X_train,
    y_train,
    epochs=10,
    batch_size=64,
    validation_data=(X_test, y_test)
)

# 9. Evaluate model
test_loss, test_acc = model.evaluate(X_test, y_test)
print("Test accuracy:", test_acc)

# 10. Plot Accuracy Curve
plt.figure()
plt.plot(history.history['accuracy'])
plt.plot(history.history['val_accuracy'])
plt.title("Model Accuracy")
plt.xlabel("Epoch")
plt.ylabel("Accuracy")
plt.legend(["Train", "Validation"])
plt.show()

# 11. Plot Loss Curve
plt.figure()
plt.plot(history.history['loss'])
plt.plot(history.history['val_loss'])
```

```
plt.title("Model Loss")
plt.xlabel("Epoch")
plt.ylabel("Loss")
plt.legend(["Train", "Validation"])
plt.show()
```

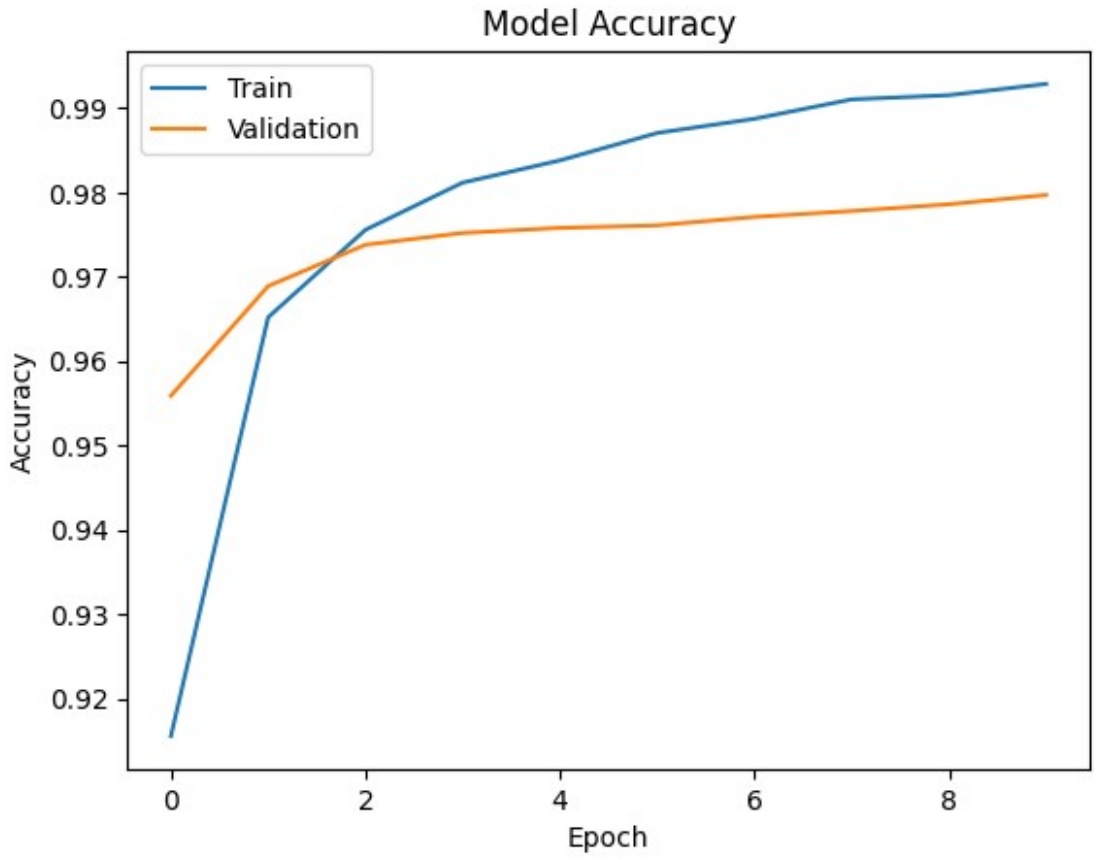
```
Model: "sequential_37"
```

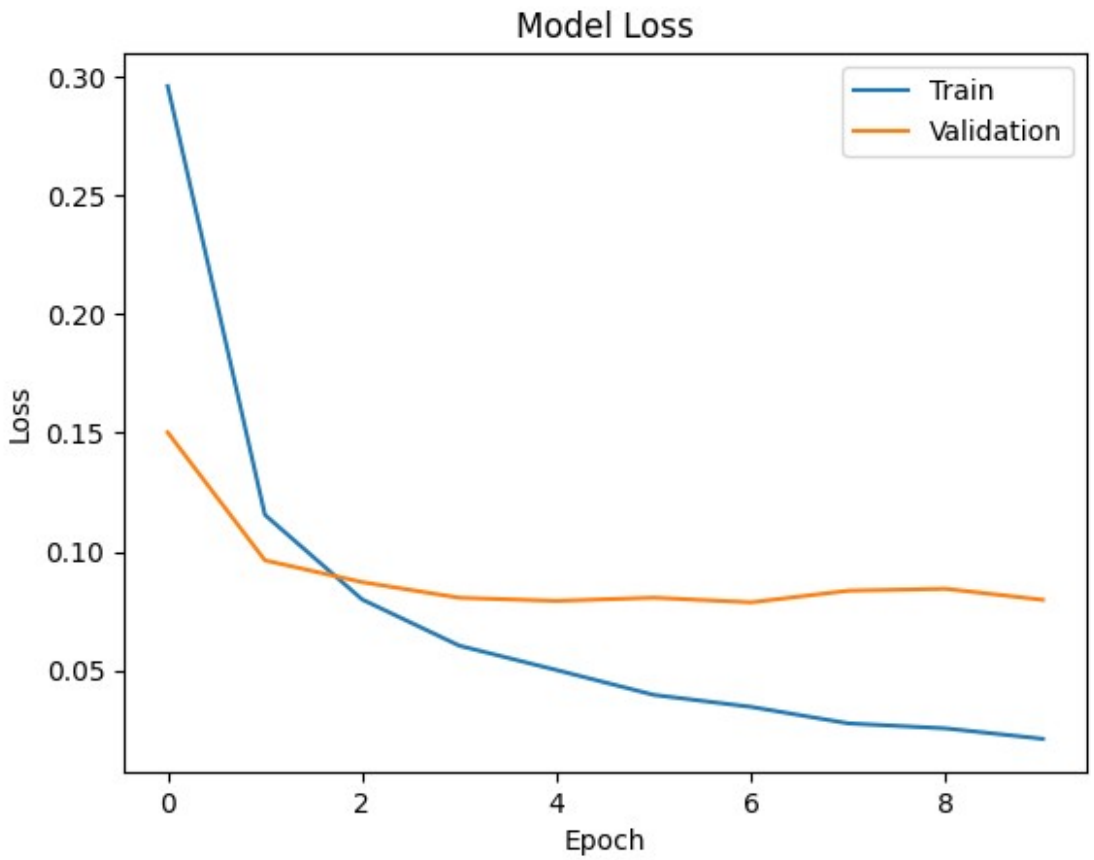
```
┏━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━┳━━━━━━━━━━━━━━━━━━━━━━━━┳━━━━━━━━━━━━━━━┓
┃ Layer (type)                    ┃ Output Shape           ┃       Param # ┃
┡━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━╇━━━━━━━━━━━━━━━━━━━━━━━━╇━━━━━━━━━━━━━━━┩
│ dense_40 (Dense)                │ (None, 128)            │       100,480 │
├─────────────────────────────────┼────────────────────────┼───────────────┤
│ dense_41 (Dense)                │ (None, 64)             │         8,256 │
├─────────────────────────────────┼────────────────────────┼───────────────┤
│ dense_42 (Dense)                │ (None, 32)             │         2,080 │
├─────────────────────────────────┼────────────────────────┼───────────────┤
│ dense_43 (Dense)                │ (None, 10)             │           330 │
└─────────────────────────────────┴────────────────────────┴───────────────┘
```

```
 Total params: 111,146 (434.16 KB)

 Trainable params: 111,146 (434.16 KB)

 Non-trainable params: 0 (0.00 B)
```

```
Epoch 1/10
938/938 ━━━━━━━━━━━━━━━━━━━━ 11s 7ms/step - accuracy: 0.8427 - loss:
0.5522 - val_accuracy: 0.9559 - val_loss: 0.1502
Epoch 2/10
938/938 ━━━━━━━━━━━━━━━━━━━━ 4s 5ms/step - accuracy: 0.9631 - loss:
0.1231 - val_accuracy: 0.9689 - val_loss: 0.0963
Epoch 3/10
938/938 ━━━━━━━━━━━━━━━━━━━━ 4s 4ms/step - accuracy: 0.9753 - loss:
0.0816 - val_accuracy: 0.9738 - val_loss: 0.0871
Epoch 4/10
938/938 ━━━━━━━━━━━━━━━━━━━━ 6s 6ms/step - accuracy: 0.9822 - loss:
0.0567 - val_accuracy: 0.9752 - val_loss: 0.0806
Epoch 5/10
938/938 ━━━━━━━━━━━━━━━━━━━━ 4s 4ms/step - accuracy: 0.9843 - loss:
```

```
0.0466 - val_accuracy: 0.9758 - val_loss: 0.0792
Epoch 6/10
938/938 ━━━━━━━━━━━━━━━━━━━━ 6s 6ms/step - accuracy: 0.9882 - loss:
0.0369 - val_accuracy: 0.9761 - val_loss: 0.0806
Epoch 7/10
938/938 ━━━━━━━━━━━━━━━━━━━━ 9s 4ms/step - accuracy: 0.9893 - loss:
0.0316 - val_accuracy: 0.9771 - val_loss: 0.0786
Epoch 8/10
938/938 ━━━━━━━━━━━━━━━━━━━━ 6s 6ms/step - accuracy: 0.9914 - loss:
0.0263 - val_accuracy: 0.9778 - val_loss: 0.0835
Epoch 9/10
938/938 ━━━━━━━━━━━━━━━━━━━━ 4s 4ms/step - accuracy: 0.9925 - loss:
0.0220 - val_accuracy: 0.9786 - val_loss: 0.0843
Epoch 10/10
938/938 ━━━━━━━━━━━━━━━━━━━━ 4s 5ms/step - accuracy: 0.9933 - loss:
0.0201 - val_accuracy: 0.9797 - val_loss: 0.0797
313/313 ━━━━━━━━━━━━━━━━━━━━ 1s 2ms/step - accuracy: 0.9766 - loss:
0.0913
Test accuracy: 0.9797000288963318
```






```
import tensorflow as tf
from tensorflow.keras.datasets import mnist
from tensorflow.keras.utils import to_categorical
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense
import matplotlib.pyplot as plt
import numpy as np
from PIL import Image
from google.colab import files

# 1. Load MNIST dataset
(X_train, y_train), (X_test, y_test) = mnist.load_data()

# 2. Normalize pixel values
X_train = X_train / 255.0
X_test = X_test / 255.0

# 3. Flatten images (28x28 → 784)
X_train = X_train.reshape(-1, 784)
X_test = X_test.reshape(-1, 784)

# 4. One-hot encoding
y_train = to_categorical(y_train, 10)
```

```
y_test = to_categorical(y_test, 10)

# 5. Build model
model = Sequential([
    Dense(128, activation='relu', input_shape=(784,)),
    Dense(64, activation='relu'),
    Dense(32, activation='relu'),
    Dense(10, activation='softmax')
])

# 6. Compile model
model.compile(
    optimizer='adam',
    loss='categorical_crossentropy',
    metrics=['accuracy']
)

# 7. Train model
history = model.fit(
    X_train,
    y_train,
    epochs=10,
    batch_size=32,
    validation_data=(X_test, y_test)
)

# 8. Evaluate model
test_loss, test_acc = model.evaluate(X_test, y_test)
print("Test accuracy:", test_acc)

# 9. Upload handwritten image
print("Upload your handwritten digit image")
uploaded = files.upload()

# 10. Load image
filename = list(uploaded.keys())[0]
img = Image.open(filename).convert("L")

# 11. Resize to 28x28
img = img.resize((28,28))

# 12. Convert to numpy
img_array = np.array(img)

# 13. Invert colors (important for MNIST)
img_array = 255 - img_array

# 14. Normalize
img_array = img_array / 255.0
```

```python
# 15. Flatten
img_array = img_array.reshape(1,784)

# 16. Predict digit
prediction = model.predict(img_array)
digit = np.argmax(prediction)

print("Predicted Digit:", digit)

# 17. Display image
plt.imshow(img_array.reshape(28,28), cmap='gray')
plt.title(f"Predicted Digit: {digit}")
plt.axis('off')
plt.show()
```

```
Epoch 1/10
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 9s 4ms/step - accuracy: 0.8569 - loss:
0.4677 - val_accuracy: 0.9625 - val_loss: 0.1290
Epoch 2/10
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 7s 4ms/step - accuracy: 0.9668 - loss:
0.1119 - val_accuracy: 0.9698 - val_loss: 0.0991
Epoch 3/10
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 9s 5ms/step - accuracy: 0.9767 - loss:
0.0750 - val_accuracy: 0.9725 - val_loss: 0.0910
Epoch 4/10
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 7s 4ms/step - accuracy: 0.9816 - loss:
0.0580 - val_accuracy: 0.9734 - val_loss: 0.0858
Epoch 5/10
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 8s 4ms/step - accuracy: 0.9854 - loss:
0.0438 - val_accuracy: 0.9763 - val_loss: 0.0803
Epoch 6/10
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 8s 4ms/step - accuracy: 0.9889 - loss:
0.0344 - val_accuracy: 0.9751 - val_loss: 0.0903
Epoch 7/10
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 7s 4ms/step - accuracy: 0.9899 - loss:
0.0302 - val_accuracy: 0.9755 - val_loss: 0.0987
Epoch 8/10
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 9s 5ms/step - accuracy: 0.9922 - loss:
0.0237 - val_accuracy: 0.9751 - val_loss: 0.0947
Epoch 9/10
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 7s 4ms/step - accuracy: 0.9925 - loss:
0.0223 - val_accuracy: 0.9726 - val_loss: 0.1079
Epoch 10/10
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 8s 4ms/step - accuracy: 0.9938 - loss:
0.0192 - val_accuracy: 0.9743 - val_loss: 0.1183
313/313 ━━━━━━━━━━━━━━━━━━━━ 1s 2ms/step - accuracy: 0.9693 - loss:
0.1500
Test accuracy: 0.9743000268936157
Upload your handwritten digit image
```